



Green Computer Vision: Comparing Accuracy and Efficiency in Deep Image Classification



1. Motivation



Deep learning drives powerful image classification but can be energy-intensive and costly to deploy.



We compare lightweight and heavier CNN architectures to evaluate accuracy, latency, energy use, and model efficiency.



Goal: Identify models that deliver high performance with low resource consumption for sustainable, real-world applications.

2. Datasets



PlantDoc Dataset
Plant Disease classification



-25,000 images
13 plant species
26 disease classes
(RGB, field captured)



TrashNet Dataset
Waste classification



-2,527 images
6 waste categories (glass, metal, paper, plastic, cardboard, trash)
(RGB)

3. Methodology Pipeline



Data collection



Preprocessing & augmentation



Model training



Evaluation (accuracy, latency, energy, CO₂e)



Sustainability analysis



Insights & recommendation

4. Research Questions

RQ1

How do CNN architectures compare in classification performance on PlantDoc and TrashNet?

RQ2

What are the trade-offs between accuracy, latency, and efficiency?

RQ3

How much inference energy is consumed by different models?

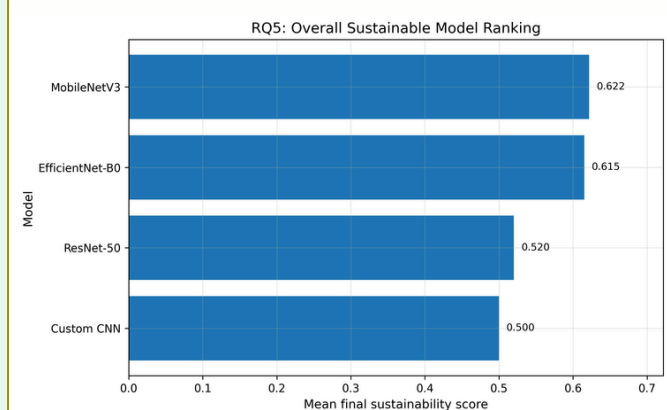
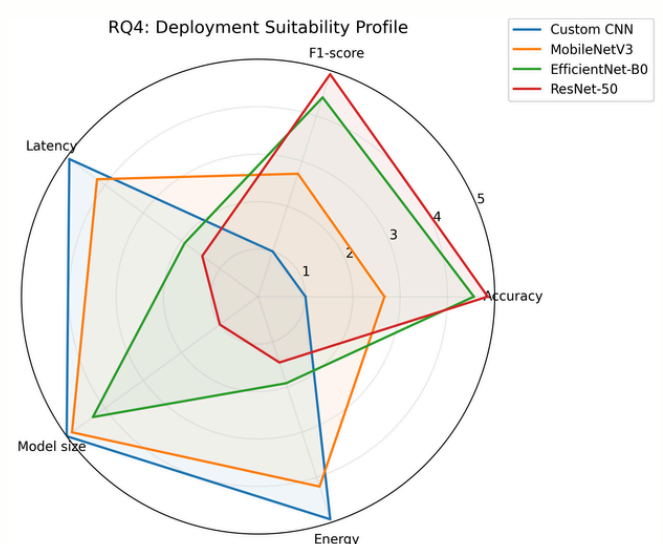
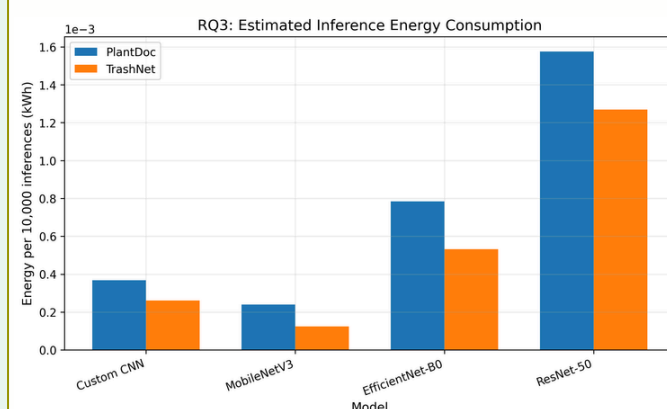
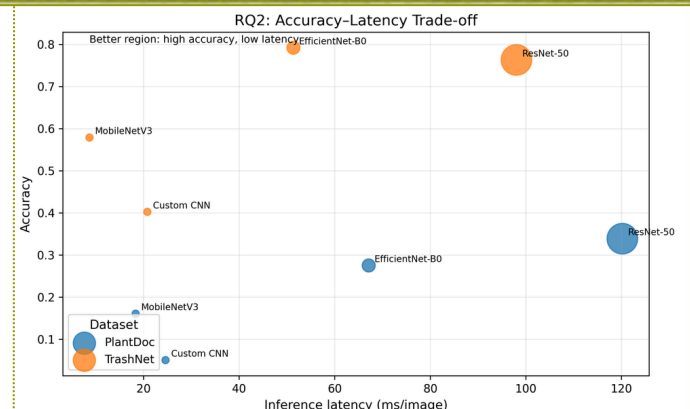
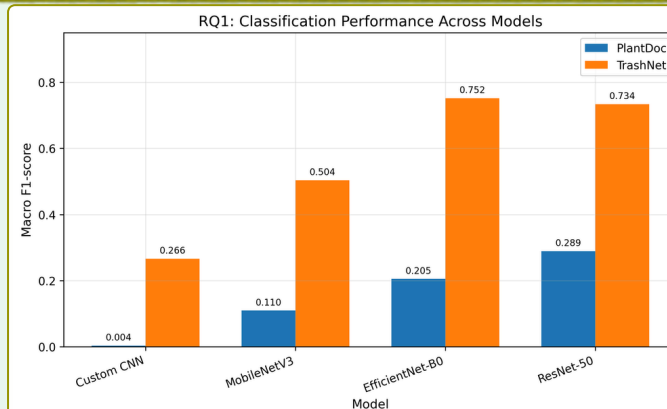
RQ4

Which models are most suitable for deployment on resource-constrained devices?

RQ5

Which model gives the best overall sustainability ranking?

5. Key Results



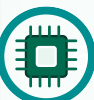
6. Contributions



Comprehensive comparison of four CNN architectures on two sustainability-focused datasets: PlantDoc and TrashNet.



Joint evaluation across accuracy, latency, inference energy, and deployment suitability.



Multi-criteria sustainability ranking to guide model selection for real-world deployment.



Energy and carbon footprint analysis for inference at scale.



Reproducible workflow with code, models, and evaluation pipeline.

7. Conclusion



Our results show that MobileNetV3 and EfficientNet-B0 offer the best balance of accuracy and efficiency across both PlantDoc and TrashNet. They achieve strong performance while maintaining moderate latency and energy consumption. ResNet-50 delivers the highest accuracy but at a substantially higher cost in latency, energy, and model size. Custom CNN is lightweight and fast but lags significantly in accuracy.

Takeaways

- Choose MobileNetV3 or EfficientNet-B0 for sustainable, real-time applications.
- ResNet-50 is suitable when maximum accuracy is critical and resources are ample.
- Lightweight models enable deployment on edge and resource-constrained devices.
- Energy-aware model selection reduces operational cost and carbon footprint.
- Sustainability metrics are essential for responsible AI deployment.

